

ANSI vs non-ANSI ports

You have seen module boundaries and port directions, now the question is where those directions are written

Two header styles

- In non-ANSI style
- In ANSI style, the header carries the full picture
- For a clocked output you might write output reg q in ANSI
- Bus widths and parameters follow the same pattern

Browser lab

igital Design and Verification Platform

HomeToolsANSI vs non-ANSI

Challenges (1/22)

Quiz: year. Non-ANSI style is associated with Verilog? Answer: 1995

Your answer

Show hintCheckNextIdle

quiz-1995

quiz-2001

quiz-names

quiz-header

starter

non-has-body-input

ansi-has-input-wire

preset-reg

reg-split

preset-bus

preset-param

preset-inout

explain

quiz-inout-net

variant-click

quiz-same

non-module-list

quiz-output-reg

param-body

quiz-prefer

inout-z

full-path

Core ideas

1995 non-ANSI

Port list = names. Directions declared in the body.

2001 ANSI

Direction, type, and width in the header.

Same RTL

Style only — functionality and hierarchy match.

Variants

Pick an example — both columns stay in sync.

Combo AND (wire out)

Clocked (output reg)

Bus widths

Parameter + ports

inout port

Continuous assign — output is a net in both styles.

• Non-ANSI port list is names only: (a, b, y)

• Directions live in the body: input / output

• ANSI puts direction (+ type) in the header

Preset combo AND

Preset output reg FF

Preset 8-bit buses

Preset parameter

Preset inout

Explain difference

Side by side

1995 non-ANSI

module and2(a, b, y);
input a, b;
output y;
assign y = a & b;
endmodule

2001 ANSI

module and2(
input wire a,
input wire b,
output wire y
);
assign y = a & b;
endmodule

(pick a variant or explain)

starter: combo AND

Cheat sheet

Topic	Rule
non-ANSI list	module m(a, b, y) — identifiers only
ANSI list	input wire a, ... inside ()
output reg	ANSI: one decl, 1995: output + reg
inout	Always a net — never reg
Modern SV	Prefer ANSI + logic (later labs)

• Starter AND: body `input a, b; output y;` -- header directions.

• Order of names in the non-ANSI list must match connections / docs.

ANSI ports lab starter

learn_verilog — ansi ports

Real Verilog practice

```
wsl - module04-ansi-ports/examples/navigation

# Track A - real Unix

# real Verilog session (Track A)

$ cat examples/ansi_ports/and2_styles.v
// and2_styles.v - same AND, two port-list conventions
module and2_nonansi(a, b, y);
    input  a, b;
    output y;
    assign y = a & b;
endmodule

module and2_ansi(
    input  wire a,
    input  wire b,
    output wire y
);
    assign y = a & b;
endmodule

$ iverilog -t null examples/ansi_ports/and2_styles.v
(syntax check passed)
```

Pitfalls to watch

- Do not assume non-ANSI means wrong, it is still valid IEEE 1364, just older layout
- Do not forget that output reg in ANSI replaces the separate output and reg lines in the
- Mixing styles in one module is confusing, pick one convention per file
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter and finish a few challenges
- On real Verilog
- When you are ready, take the short quiz, then continue to operators in the next module